FastAPI Reference OpenAPI

OpenAPI `docs`

Utilities to handle OpenAPI automatic UI documentation, including Swagger UI (by default at `/docs`) and ReDoc (by default at `/redoc`).

`func` fastapi.openapi.docs.get_swagger_ui_html

```
get_swagger_ui_html(
    *,
    openapi_url,
    title,
    swagger_js_url="https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui-bundle.js",
    swagger_css_url="https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui.css",
    swagger_favicon_url="https://fastapi.tiangolo.com/img/favicon.png",
    oauth2_redirect_url=None,
    init_oauth=None,
    swagger_ui_parameters=None
)
```

Generate and return the HTML that loads Swagger UI for the interactive API docs (normally served at `/docs`).

You would only call this function yourself if you needed to override some parts, for example the URLs to use to load Swagger UI's JavaScript and CSS.

Read more about it in the FastAPI docs for Configure Swagger UI and the FastAPI docs for Custom Docs UI Static Assets (Self-Hosting).

PARAMETER	DESCRIPTION
`openapi_url`	The OpenAPI URL that Swagger UI should load and use. This is normally done automatically by FastAPI using the default URL `/openapi.json`. Read more about it in the FastAPI docs for Conditional OpenAPI **TYPE:** `str`
`title`	The HTML `<title>` content, normally shown in the browser tab. Read more about it in the FastAPI docs for Custom Docs UI Static Assets **TYPE:** `str`
`swagger_js_url`	The URL to use to load the Swagger UI JavaScript. It is normally set to a CDN URL. Read more about it in the FastAPI docs for Custom Docs UI Static Assets **TYPE:** `str` **DEFAULT:** `'https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui-bundle.js'`

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PARAMETER	DESCRIPTION
`swagger_css_url`	The URL to use to load the Swagger UI CSS. It is normally set to a CDN URL. Read more about it in the FastAPI docs for Custom Docs UI Static Assets **TYPE:** `str` **DEFAULT:** `'https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui.css'`
`swagger_favicon_url`	The URL of the favicon to use. It is normally shown in the browser tab. **TYPE:** `str` **DEFAULT:** `'https://fastapi.tiangolo.com/img/favicon.png'`
`oauth2_redirect_url`	The OAuth 2 redirect URL, it is normally automatically handled by FastAPI. Read more about it in the FastAPI docs for Custom Docs UI Static Assets **TYPE:** `str | None` **DEFAULT:** `None`
`init_oauth`	A dictionary with Swagger UI OAuth 2 initialization configurations. Read more about the available configuration options in the Swagger UI docs. **TYPE:** `dict[str, Any] | None` **DEFAULT:** `None`
`swagger_ui_parameters`	Configuration parameters for Swagger UI. It defaults to swagger_ui_default_parameters. Read more about it in the FastAPI docs about how to Configure Swagger UI. **TYPE:** `dict[str, Any] | None` **DEFAULT:** `None`

Source code in `fastapi/openapi/docs.py`

```python
def get_swagger_ui_html(
    *,
    openapi_url: Annotated[
        str,
        Doc(
            """
            The OpenAPI URL that Swagger UI should load and use.

            This is normally done automatically by FastAPI using the default URL
            `/openapi.json`.

            Read more about it in the
            [FastAPI docs for Conditional OpenAPI](https://fastapi.tiangolo.com/how-to/conditional-openapi/#conditional-openapi-from-settings-and-env-vars)
            """
        ),
    ],
    title: Annotated[
        str,
        Doc(
            """
            The HTML `<title>` content, normally shown in the browser tab.

            Read more about it in the
            [FastAPI docs for Custom Docs UI Static Assets](https://fastapi.tiangolo.com/how-to/custom-docs-ui-assets/)
            """
        ),
    ],
    swagger_js_url: Annotated[
        str,
        Doc(
            """
            The URL to use to load the Swagger UI JavaScript.

            It is normally set to a CDN URL.

            Read more about it in the
            [FastAPI docs for Custom Docs UI Static Assets](https://fastapi.tiangolo.com/how-to/custom-docs-ui-assets/)
            """
        ),
    ] = "https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui-bundle.js",
    swagger_css_url: Annotated[
        str,
        Doc(
            """
            The URL to use to load the Swagger UI CSS.

            It is normally set to a CDN URL.

            Read more about it in the
            [FastAPI docs for Custom Docs UI Static Assets](https://fastapi.tiangolo.com/how-to/custom-docs-ui-assets/)
            """
        ),
    ] = "https://cdn.jsdelivr.net/npm/swagger-ui-dist@5/swagger-ui.css",
    swagger_favicon_url: Annotated[
        str,
        Doc(
            """
            The URL of the favicon to use. It is normally shown in the browser tab.
            """
        ),
    ] = "https://fastapi.tiangolo.com/img/favicon.png",
    oauth2_redirect_url: Annotated[
        str | None,
        Doc(
            """
            The OAuth2 redirect URL, it is normally automatically handled by FastAPI.

            Read more about it in the
```

```
            [FastAPI docs for Custom Docs UI Static Assets](https://fastapi.tiangolo.com/how-
to/custom-docs-ui-assets/)
            """
        ),
    ] = None,
    init_oauth: Annotated[
        dict[str, Any] | None,
        Doc(
            """
            A dictionary with Swagger UI OAuth2 initialization configurations.

            Read more about the available configuration options in the
            [Swagger UI docs](https://swagger.io/docs/open-source-tools/swagger-ui/usage/oauth2/).
            """
        ),
    ] = None,
    swagger_ui_parameters: Annotated[
        dict[str, Any] | None,
        Doc(
            """
            Configuration parameters for Swagger UI.

            It defaults to [swagger_ui_default_parameters]
[fastapi.openapi.docs.swagger_ui_default_parameters].

            Read more about it in the
            [FastAPI docs about how to Configure Swagger UI](https://fastapi.tiangolo.com/how-
to/configure-swagger-ui/).
            """
        ),
    ] = None,
) -> HTMLResponse:
    """
    Generate and return the HTML  that loads Swagger UI for the interactive
    API docs (normally served at `/docs`).

    You would only call this function yourself if you needed to override some parts,
    for example the URLs to use to load Swagger UI's JavaScript and CSS.

    Read more about it in the
    [FastAPI docs for Configure Swagger UI](https://fastapi.tiangolo.com/how-to/configure-swagger-
ui/)
    and the [FastAPI docs for Custom Docs UI Static Assets (Self-Hosting)]
(https://fastapi.tiangolo.com/how-to/custom-docs-ui-assets/).
    """
    current_swagger_ui_parameters = swagger_ui_default_parameters.copy()
    if swagger_ui_parameters:
        current_swagger_ui_parameters.update(swagger_ui_parameters)

    html = f"""
    <!DOCTYPE html>
    <html>
    <head>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <link type="text/css" rel="stylesheet" href="{swagger_css_url}">
    <link rel="shortcut icon" href="{swagger_favicon_url}">
    <title>{title}</title>
    </head>
    <body>
    <div id="swagger-ui">
    </div>
    <script src="{swagger_js_url}"></script>
    <!-- `SwaggerUIBundle` is now available on the page -->
    <script>
    const ui = SwaggerUIBundle({{
        url: '{openapi_url}',
    """

    for key, value in current_swagger_ui_parameters.items():
        html += f"{_html_safe_json(key)}: {_html_safe_json(jsonable_encoder(value))},\n"

    if oauth2_redirect_url:
        html += f"oauth2RedirectUrl: window.location.origin + '{oauth2_redirect_url}',"

    html += """
    presets: [
```

```
        SwaggerUIBundle.presets.apis,
        SwaggerUIBundle.SwaggerUIStandalonePreset
        ],
    })"""

    if init_oauth:
        html += f"""
        ui.initOAuth({_html_safe_json(jsonable_encoder(init_oauth))})
        """

    html += """
    </script>
    </body>
    </html>
    """
    return HTMLResponse(html)
```

func fastapi.openapi.docs.get_redoc_html

```
get_redoc_html(
    *,
    openapi_url,
    title,
    redoc_js_url="https://cdn.jsdelivr.net/npm/redoc@2/bundles/redoc.standalone.js",
    redoc_favicon_url="https://fastapi.tiangolo.com/img/favicon.png",
    with_google_fonts=True
)
```

Generate and return the HTML response that loads ReDoc for the alternative API docs (normally served at `/redoc`).

You would only call this function yourself if you needed to override some parts, for example the URLs to use to load ReDoc's JavaScript and CSS.

Read more about it in the FastAPI docs for Custom Docs UI Static Assets (Self-Hosting).

PARAMETER	DESCRIPTION
`openapi_url`	The OpenAPI URL that ReDoc should load and use. This is normally done automatically by FastAPI using the default URL `/openapi.json`. Read more about it in the FastAPI docs for Conditional OpenAPI **TYPE:** `str`
`title`	The HTML `<title>` content, normally shown in the browser tab. Read more about it in the FastAPI docs for Custom Docs UI Static Assets **TYPE:** `str`
`redoc_js_url`	The URL to use to load the ReDoc JavaScript. It is normally set to a CDN URL. Read more about it in the FastAPI docs for Custom Docs UI Static Assets **TYPE:** `str` **DEFAULT:** `'https://cdn.jsdelivr.net/npm/redoc@2/bundles/redoc.standalone.js'`

PARAMETER	DESCRIPTION
`redoc_favicon_url`	The URL of the favicon to use. It is normally shown in the browser tab. **TYPE:** `str` **DEFAULT:** `'https://fastapi.tiangolo.com/img/favicon.png'`
`with_google_fonts`	Load and use Google Fonts. **TYPE:** `bool` **DEFAULT:** `True`

Source code in `fastapi/openapi/docs.py`

```python
def get_redoc_html(
    *,
    openapi_url: Annotated[
        str,
        Doc(
            """
            The OpenAPI URL that ReDoc should load and use.

            This is normally done automatically by FastAPI using the default URL
            `/openapi.json`.

            Read more about it in the
            [FastAPI docs for Conditional OpenAPI](https://fastapi.tiangolo.com/how-to/conditional-openapi/#conditional-openapi-from-settings-and-env-vars)
            """
        ),
    ],
    title: Annotated[
        str,
        Doc(
            """
            The HTML `<title>` content, normally shown in the browser tab.

            Read more about it in the
            [FastAPI docs for Custom Docs UI Static Assets](https://fastapi.tiangolo.com/how-to/custom-docs-ui-assets/)
            """
        ),
    ],
    redoc_js_url: Annotated[
        str,
        Doc(
            """
            The URL to use to load the ReDoc JavaScript.

            It is normally set to a CDN URL.

            Read more about it in the
            [FastAPI docs for Custom Docs UI Static Assets](https://fastapi.tiangolo.com/how-to/custom-docs-ui-assets/)
            """
        ),
    ] = "https://cdn.jsdelivr.net/npm/redoc@2/bundles/redoc.standalone.js",
    redoc_favicon_url: Annotated[
        str,
        Doc(
            """
            The URL of the favicon to use. It is normally shown in the browser tab.
            """
        ),
    ] = "https://fastapi.tiangolo.com/img/favicon.png",
    with_google_fonts: Annotated[
        bool,
        Doc(
            """
            Load and use Google Fonts.
            """
        ),
    ] = True,
) -> HTMLResponse:
    """
    Generate and return the HTML response that loads ReDoc for the alternative
    API docs (normally served at `/redoc`).

    You would only call this function yourself if you needed to override some parts,
    for example the URLs to use to load ReDoc's JavaScript and CSS.

    Read more about it in the
    [FastAPI docs for Custom Docs UI Static Assets (Self-Hosting)](https://fastapi.tiangolo.com/how-to/custom-docs-ui-assets/).
    """
    html = f"""
```

```
    <!DOCTYPE html>
    <html>
    <head>
    <title>{title}</title>
    <!-- needed for adaptive design -->
    <meta charset="utf-8"/>
    <meta name="viewport" content="width=device-width, initial-scale=1">
    """
    if with_google_fonts:
        html += """
    <link href="https://fonts.googleapis.com/css?family=Montserrat:300,400,700|Roboto:300,400,700"
rel="stylesheet">
    """
    html += f"""
    <link rel="shortcut icon" href="{redoc_favicon_url}">
    <!--
    ReDoc doesn't change outer page styles
    -->
    <style>
      body {{
        margin: 0;
        padding: 0;
      }}
    </style>
    </head>
    <body>
    <noscript>
        ReDoc requires Javascript to function. Please enable it to browse the documentation.
    </noscript>
    <redoc spec-url="{openapi_url}"></redoc>
    <script src="{redoc_js_url}"> </script>
    </body>
    </html>
    """
    return HTMLResponse(html)
```

`func` fastapi.openapi.docs.get_swagger_ui_oauth2_redirect_html

```
get_swagger_ui_oauth2_redirect_html()
```

Generate the HTML response with the OAuth2 redirection for Swagger UI.

You normally don't need to use or change this.

Source code in `fastapi/openapi/docs.py`

```python
def get_swagger_ui_oauth2_redirect_html() -> HTMLResponse:
    """
    Generate the HTML response with the OAuth2 redirection for Swagger UI.

    You normally don't need to use or change this.
    """
    # copied from https://github.com/swagger-api/swagger-ui/blob/v4.14.0/dist/oauth2-redirect.html
    html = """
    <!doctype html>
    <html lang="en-US">
    <head>
        <title>Swagger UI: OAuth2 Redirect</title>
    </head>
    <body>
    <script>
        'use strict';
        function run () {
            var oauth2 = window.opener.swaggerUIRedirectOauth2;
            var sentState = oauth2.state;
            var redirectUrl = oauth2.redirectUrl;
            var isValid, qp, arr;

            if (/code|token|error/.test(window.location.hash)) {
                qp = window.location.hash.substring(1).replace('?', '&');
            } else {
                qp = location.search.substring(1);
            }

            arr = qp.split("&");
            arr.forEach(function (v,i,_arr) { _arr[i] = '"' + v.replace('=', '":"') + '"';});
            qp = qp ? JSON.parse('{' + arr.join() + '}',
                    function (key, value) {
                        return key === "" ? value : decodeURIComponent(value);
                    }
            ) : {};

            isValid = qp.state === sentState;

            if ((
              oauth2.auth.schema.get("flow") === "accessCode" ||
              oauth2.auth.schema.get("flow") === "authorizationCode" ||
              oauth2.auth.schema.get("flow") === "authorization_code"
            ) && !oauth2.auth.code) {
                if (!isValid) {
                    oauth2.errCb({
                        authId: oauth2.auth.name,
                        source: "auth",
                        level: "warning",
                        message: "Authorization may be unsafe, passed state was changed in server.
The passed state wasn't returned from auth server."
                    });
                }

                if (qp.code) {
                    delete oauth2.state;
                    oauth2.auth.code = qp.code;
                    oauth2.callback({auth: oauth2.auth, redirectUrl: redirectUrl});
                } else {
                    let oauthErrorMsg;
                    if (qp.error) {
                        oauthErrorMsg = "["+qp.error+"]: " +
                            (qp.error_description ? qp.error_description+ ". " : "no accessCode
received from the server. ") +
                            (qp.error_uri ? "More info: "+qp.error_uri : "");
                    }

                    oauth2.errCb({
                        authId: oauth2.auth.name,
                        source: "auth",
                        level: "error",
                        message: oauthErrorMsg || "[Authorization failed]: no accessCode received
from the server."
```

```
                    });
                }
            } else {
                oauth2.callback({auth: oauth2.auth, token: qp, isValid: isValid, redirectUrl:
redirectUrl});
            }
            window.close();
        }

        if (document.readyState !== 'loading') {
            run();
        } else {
            document.addEventListener('DOMContentLoaded', function () {
                run();
            });
        }
    </script>
    </body>
    </html>
        """
    return HTMLResponse(content=html)
```

attr fastapi.openapi.docs.swagger_ui_default_parameters module-attribute

```
swagger_ui_default_parameters = {
    "dom_id": "#swagger-ui",
    "layout": "BaseLayout",
    "deepLinking": True,
    "showExtensions": True,
    "showCommonExtensions": True,
}
```

Default configurations for Swagger UI.

You can use it as a template to add any other configurations needed.